

AAE 451B – A5

Team 01

9/13/26



School of Aeronautics
and Astronautics

Assumed Flight Conditions

- Velocities:
 - Cruise: 23.5 m/s
 - Stall: 10.5 m/s
 - Max: 28.2 m/s
- Selected wingspan: 1.5 m
- Chord: 0.318 m
 - From selected wing loading of 95 N/m², estimated mass of 4.45 kg, and fuselage width of 55 mm
- Air properties at standard sea level conditions: 101325 Pa, 288 K
 - Density: 1.225 kg/m³
 - Viscosity: 1.796E-05 Pa-s
- Reynolds number: Chord used as reference length
 - Cruise velocity: 510000
 - Stall velocity: 230000
 - Max velocity: 610000
- Reynolds number range used for airfoil trade: 100000 – 700000

Airfoil Trade Study – Criteria

Criterion	Requirements it serves (reference from A1)	Weight	Relative Weight	Weight Justification
$C_{l,max}$	R1	3	0.214	The flight mission includes a baseline requirement of taking off within 25 meters of the starting line. A high maximum lift coefficient is required to guarantee a successful takeoff relative to this requirement.
C_{m0}	R4	3	0.214	Highly cambered airfoils generate massive nose-down pitching moments. Balancing this requires the tail to generate significant downforce, which would result in a high level of trim drag.
$(L/D)_{max}$	R3, R21	3	0.214	The aircraft must sustain flight for 3 laps under a strict power cap. Maximizing aerodynamic efficiency directly improves the power efficiency during the cruising state.
Manufacturability	R23, R24	1	0.0714	Choosing an airfoil that does not overtly cut into our budget or is too challenging to effectively manufacture is a crucial part of the design process that must be considered.
α_{stall}	R5, R19	1	0.0714	A higher stall angle provides a forgiving margin of error for the pilot in the case of any sudden actions that may be needed in the air
$C_{l,\alpha}$	R19	1	0.0714	Having a lift curve slope in either extremity (too steep/flat) may lead to overly sensitive or sluggish flight behaviors.
$C_{d,min}$	R21	1	0.0714	Lower base drag saves battery power. It gets a lower weight than $(L/D)_{max}$ because it only shows the 'best-case' drag, which might not be the angle our aircraft flies at during cruise.
Alpha at $(L/D)_{max}$	R3	1	0.0714	Aligning the airfoil's most efficient angle of attack with the aircraft's flat cruising angle can help ensure drag is minimized during flight.

Airfoil	Cl,max	Cm0	(L/D)max	Manufacturability	Alphastall	Cl,alpha	Cd,min	Alpha @ (L/D)max	Score
AG35	3	2	1	3	1	2	2	2	2.00
Clark Y	3	1	3	3	1	2	3	3	2.35
E205	2	2	2	3	2	2	2	1	2.00
E423	3	1	1	2	3	1	1	3	1.78
MH32	1	2	3	3	1	2	3	3	2.14
NACA 0012	1	3	1	3	1	2	2	2	1.78
NACA 2212	2	3	1	3	3	2	2	1	2.07
NACA 23012	1	3	1	3	2	2	2	1	1.78
NACA 23015	1	3	1	3	3	2	1	1	1.78
NACA 2412	3	2	2	3	2	2	3	2	2.35
NACA 2415	2	2	2	3	3	3	3	2	2.28
NACA 4412	3	1	2	2	3	2	1	1	1.93
NACA 4415	3	1	1	3	3	2	1	1	1.78
NACA 4418	3	2	1	3	3	2	1	1	2.00
S1210	1	1	3	1	2	2	2	2	1.71
S1223	1	1	3	1	1	3	1	3	1.71
S3014	2	3	1	1	1	2	2	2	1.86
S3021	2	2	2	1	1	2	2	1	1.78
S4320	3	1	2	1	1	2	2	1	1.78
S7055	3	2	2	2	1	2	2	1	2.07
S8036	1	3	1	3	3	1	3	1	1.86
SD7032	3	2	1	2	2	2	2	1	1.93
SD7037	3	2	1	3	2	2	2	2	2.07
SD7062	3	2	3	3	3	2	2	2	2.57
SD7080	2	2	2	3	1	2	2	2	2.00
Weight	0.214	0.214	0.214	0.0714	0.0714	0.0714	0.0714	0.0714	

Airfoil Trade Study: Airfoils

Re = 230000 - Stall Speed

1st Pick

2nd / 3rd Pick

4th Pick

Airfoil	C _{l0}	C _{m0}	C _{la}	C _{l,max}	alpha _{stall}	(L/D) _{max}	alpha at (L/D) _{max}	C _{d,min}
AG35	0.403	-0.044	6.318	1.308	10.6	58.973	5.4	0.013
CLARK Y	0.44	-0.095	6.1	1.4	12.5	73.2	4.8	0.0101
EPPLER 205	0.25	-0.045	6.199	1.246	14.2	61.994	8.4	0.014
EPPLER E423	0.92	-0.207	5.177	1.899	18.2	51.259	3.2	0.021
MH32	0.33	-0.073	6.13	1.16	10.5	72.3	4.8	0.009
NACA 0012	0	0	6.329	1.141	12.6	39.36	6	0.014
NACA 2212	0.189	-0.032	6.289	1.326	16.6	56.974	9	0.014
NACA 23012	0.123	-0.01	6.293	1.386	14	48.74	7.8	0.014
NACA 23015	0.122	-0.009	6.229	1.472	16.2	47.718	8.8	0.015
NACA 2412	0.29	-0.061	6.54	1.33	14.2	66.6	6	0.0098
NACA 2415	0.23	-0.047	7.17	1.37	17.5	64.1	6.8	0.0108
NACA 4412	0.416	-0.094	6.163	1.469	17.4	62.383	8.6	0.015
NACA 4415	0.399	-0.09	6.013	1.501	21.6	54.668	8.2	0.016
NACA 4418	0.37	-0.085	5.664	1.531	24.6	45.822	7.6	0.017
SELIG 1210	1.05	-0.245	6.5	1.96	14.5	86.3	6.5	0.0123
SELIG S1223	1.19	-0.271	7.31	2.29	13	73.6	3.8	0.0176
SELIG S3014	0.221	-0.04	6.246	1.234	13.4	57.706	7.4	0.013
SELIG S3021	0.272	-0.052	6.213	1.263	13	61.686	7.8	0.013
SELIG S4320	0.455	-0.091	6.308	1.526	12.6	67.639	7.8	0.014
SELIG S7055	0.317	-0.063	6.101	1.272	10.6	61.653	8.6	0.014
SELIG S8036	0.153	-0.028	5.6	1.294	17.8	45.171	8.6	0.015
SD7032	0.414	-0.088	6.262	1.452	13.2	65.783	8	0.014
SD7037	0.337	-0.073	6.254	1.365	13.8	62.965	7.2	0.013
SD7062	0.43	-0.08	6.29	1.59	15.8	70	6.2	0.0113
SD7080	0.28	-0.06	6.26	1.301	12.6	60.176	7	0.013

Airfoil Trade Study: Airfoils

Re = 510000 - Cruise Speed

1st Pick

2nd / 3rd Pick

4th Pick

Airfoil	C _{l0}	C _{m0}	C _{la}	C _{l,max}	alpha _{stall} (L/D) _{max}	alpha at (L/D) _{max}	C _{d,min}
AG35	0.41	-0.044	6.404	1.349	11.8	73.804	4.8
Clark Y	0.353	-0.077	6.272	1.434	13	80.324	8.4
EPPLER 205	0.254	-0.045	6.297	1.263	15.2	78.355	7.6
EPPLER E423	0.975	-0.217	5.303	1.937	16.6	67.799	3.4
MH32	0.261	-0.055	6.316	1.236	11.2	73.762	6.2
NACA 0012	0	0	6.439	1.243	14.2	50.86	7.8
NACA 2212	0.195	-0.033	6.405	1.412	14.2	75.505	9.2
NACA 23012	0.126	-0.01	6.397	1.475	14.4	68.84	11
NACA 23015	0.126	-0.009	6.387	1.535	16.6	63.8	10.6
NACA 2412	0.218	-0.049	6.398	1.382	16.6	70.879	7
NACA 2415	0.212	-0.047	6.353	1.393	15.8	71.145	8.6
NACA 4412	0.432	-0.096	6.296	1.483	13.8	81.946	8.8
NACA 4415	0.419	-0.093	6.197	1.527	14.4	72.687	8.8
NACA 4418	0.394	-0.088	5.907	1.567	14.6	61.505	8.2
SELIG 1210	1.038	-0.243	6.293	2.033	13.4	91.949	6.4
SELIG S1223	1.176	-0.255	6.649	2.332	14.4	84.221	5.2
SELIG S3014	0.223	-0.04	6.34	1.278	12	73.244	7
SELIG S3021	0.278	-0.053	6.296	1.294	13.6	77.825	7
SELIG S4320	0.465	-0.091	6.393	1.559	11.6	88.067	8.2
SELIG S7055	0.324	-0.063	6.208	1.314	12.4	80.328	8.4
SELIG S8036	0.161	-0.028	5.803	1.256	19	60.651	8.6
SD7032	0.426	-0.089	6.358	1.566	13.6	85.3	7.8
SD7037	0.347	-0.074	6.344	1.415	12.6	80.281	6.8
SD7062	0.431	-0.081	6.395	1.655	15.8	80.73	8.8
SD7080	0.288	-0.061	6.347	1.356	12.8	76.17	6.6

Airfoil Trade Study: Airfoils

Re = 610000 - Max Speed

1st Pick

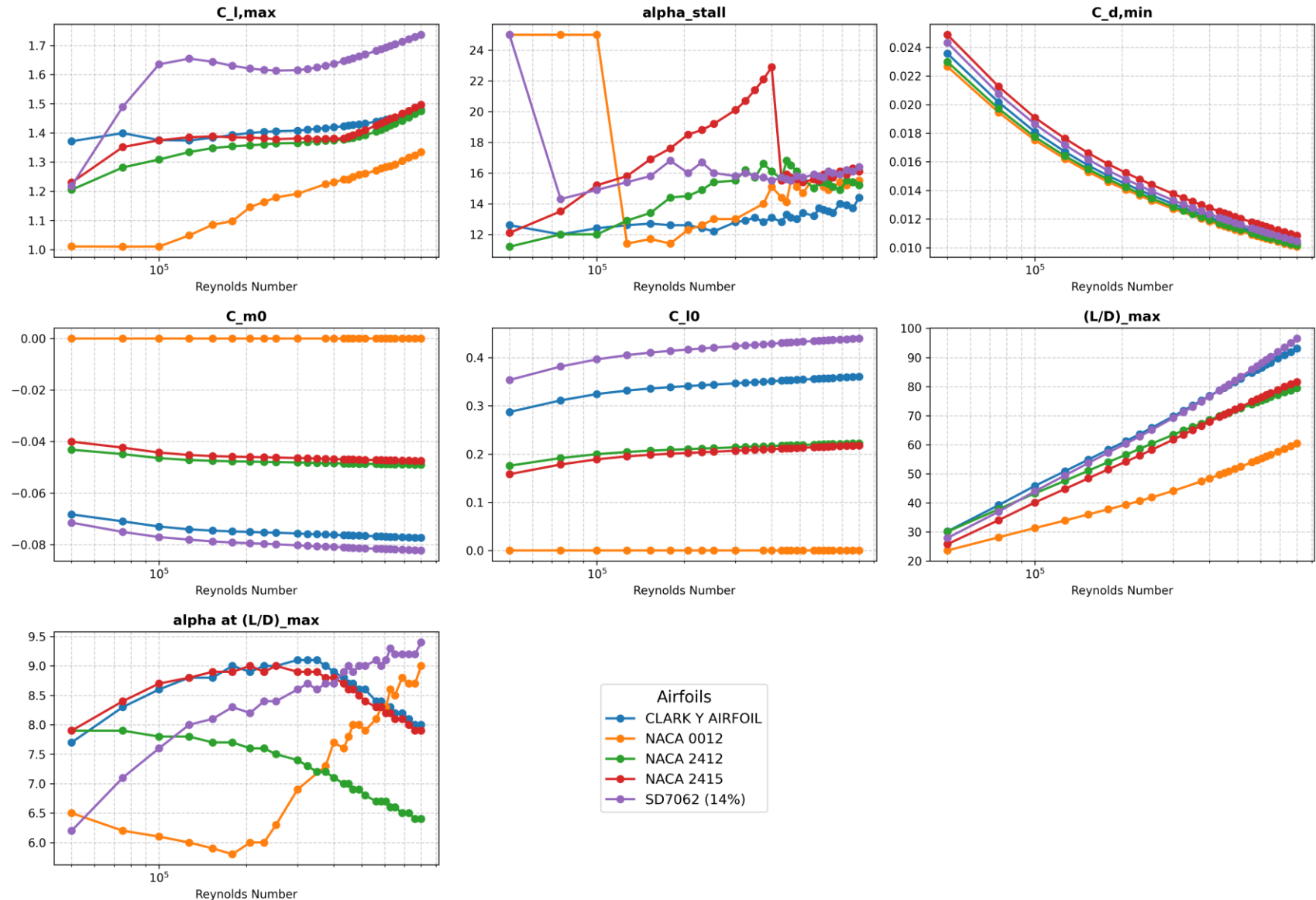
2nd / 3rd Pick

4th Pick

Airfoil	C _{l0}	C _{m0}	C _{lα}	C _{l,max}	α _{stall} (L/D) _{max}	α at (L/D) _{max}	C _{d,min}
AG35	0.411	-0.044	6.42	1.37	11.6	76.933	4.8
Clark Y	0.355	-0.077	6.297	1.436	12.8	84.631	8.2
EPPLER 205	0.255	-0.045	6.315	1.265	14.6	81.853	7.4
EPPLER E423	0.987	-0.219	5.345	1.955	16.2	72.076	3.6
MH32	0.263	-0.055	6.332	1.249	11.4	76.94	6
NACA 0012	0	0	6.46	1.267	15	53.933	8.4
NACA 2212	0.196	-0.033	6.426	1.443	14.2	79.634	9
NACA 23012	0.127	-0.01	6.416	1.496	14.8	73.585	10.8
NACA 23015	0.126	-0.009	6.419	1.566	16.6	68.694	10.6
NACA 2412	0.22	-0.049	6.421	1.404	15	73.901	6.8
NACA 2415	0.214	-0.047	6.386	1.426	15.6	74.762	8.4
NACA 4412	0.435	-0.097	6.323	1.515	14.4	86.662	8.6
NACA 4415	0.423	-0.093	6.237	1.55	14.8	77.421	9
NACA 4418	0.4	-0.088	5.975	1.583	15	65.686	8.2
SELIG 1210	1.047	-0.245	6.308	2.068	13.8	97.105	6.4
SELIG S1223	1.187	-0.258	6.626	2.35	14	88.762	5.4
SELIG S3014	0.223	-0.04	6.354	1.295	12.4	76.534	6.8
SELIG S3021	0.279	-0.053	6.312	1.305	14	81.229	6.8
SELIG S4320	0.466	-0.092	6.408	1.569	13.4	93.068	8
SELIG S7055	0.325	-0.063	6.231	1.333	12.6	84.644	8.2
SELIG S8036	0.163	-0.029	5.847	1.273	18.8	64.477	8.4
SD7032	0.429	-0.09	6.376	1.588	13.6	89.899	7.6
SD7037	0.348	-0.074	6.36	1.436	12.6	84.076	6.6
SD7062	0.434	-0.082	6.422	1.682	15.8	85.965	9
SD7080	0.289	-0.061	6.364	1.374	12.8	79.598	6.4

Airfoil Trade Study: Airfoils

Airfoil Aerodynamic Parameters vs. Reynolds Number



Wing Planform Design

- No taper, sweep, twist, or dihedral
 - Simplifies manufacturing and assembly of wing
 - Not critical to meet requirements
- Selected airfoil has slightly lower $C_{L,max}$ than assumed in A2 sizing
 - Moved stall constraint
 - Had to change design wing loading from 100 N/m² to 95 N/m²
- Wingspan selected at RFP maximum of 1.5 m
 - This maximizes AR and aerodynamic efficiency for a given wing area
- Wing area calculated from vehicle weight estimation and selected wing loading
- Chord calculated from wing area and wingspan
 - Accounted for wingspan covered by fuselage

Wing Parameters:

- Wingspan: 1.5 m
- Chord: 0.318 m
- AR: 4.72

Drag Build-Up

- Dimensions match CAD configuration
- Drag coefficients:
 - Fuselage: 0.0023
 - Wing: 0.0143
 - Horizontal stabilizer: 0.0034
 - Vertical stabilizers: 0.0026
 - Landing gear: 0.0061
 - Leakage/protuberance: 0.0029
- $CD_0 = 0.0315$
- Oswald span efficiency empirical correlation from Sadraey equation 5.27a
- $e = 0.91$

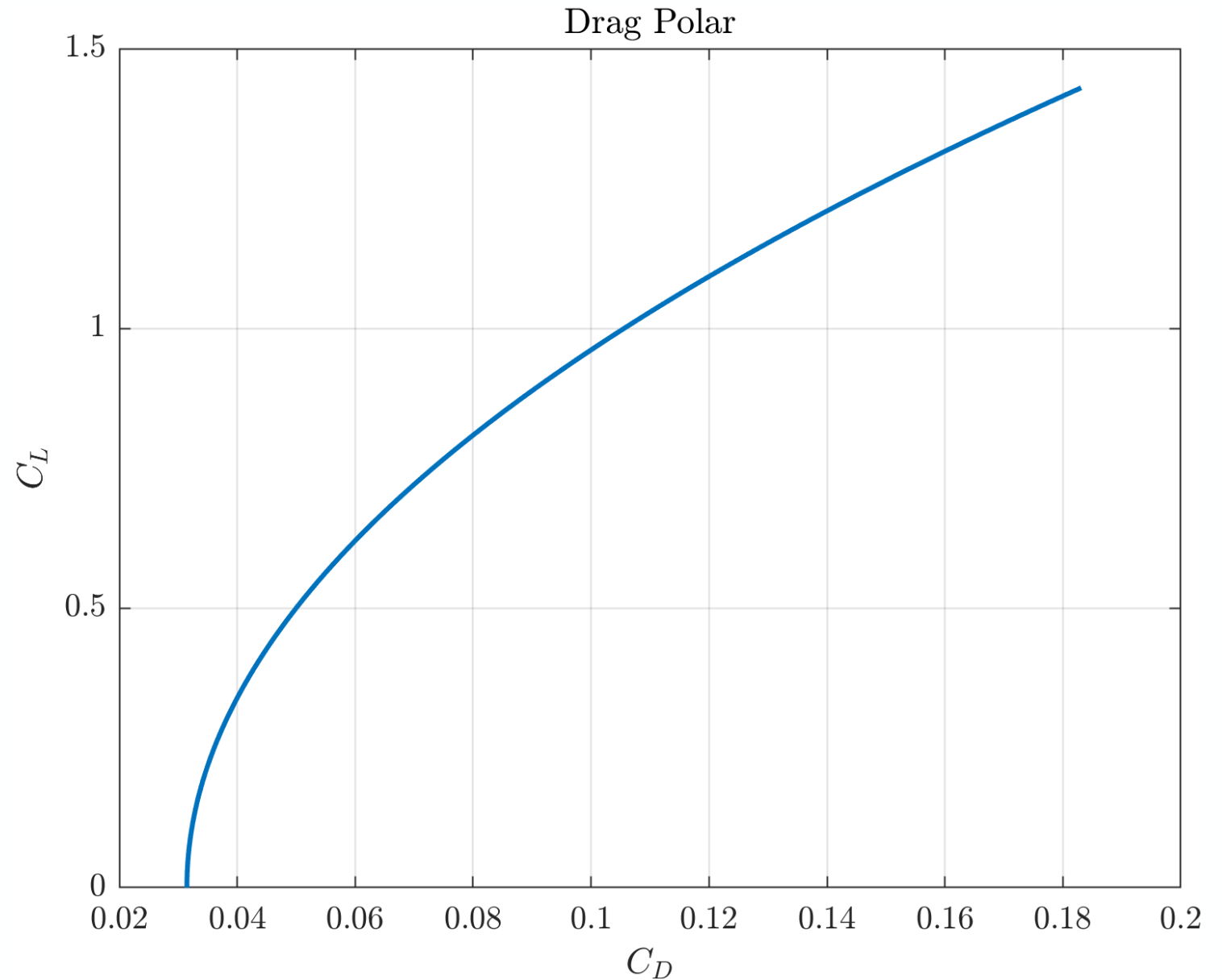
The Oswald span efficiency for a straight wing and a swept wing is given respectively by Equation (5.27a,b) [12]:

$$e = 1.78 (1 - 0.045AR^{0.68}) - 0.64 \quad (5.27a)$$

$$e = 4.61 (1 - 0.045AR^{0.68}) [\cos(\Lambda_{LE})]^{0.15} - 3.1 \quad (5.27b)$$

Clean Drag Polar

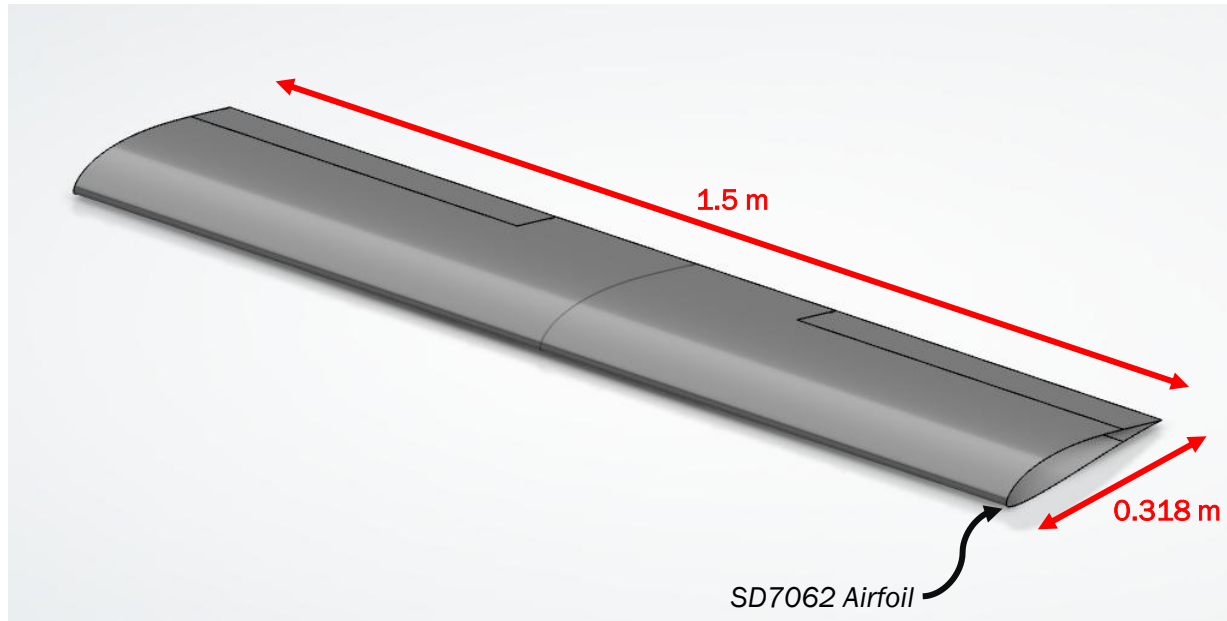
- $(L/D)_{\max} = 10.35$
 - At $C_L = 0.65$
- $(L/D)_{\text{cruise}} = 7.87$
 - At $C_L = 0.30$



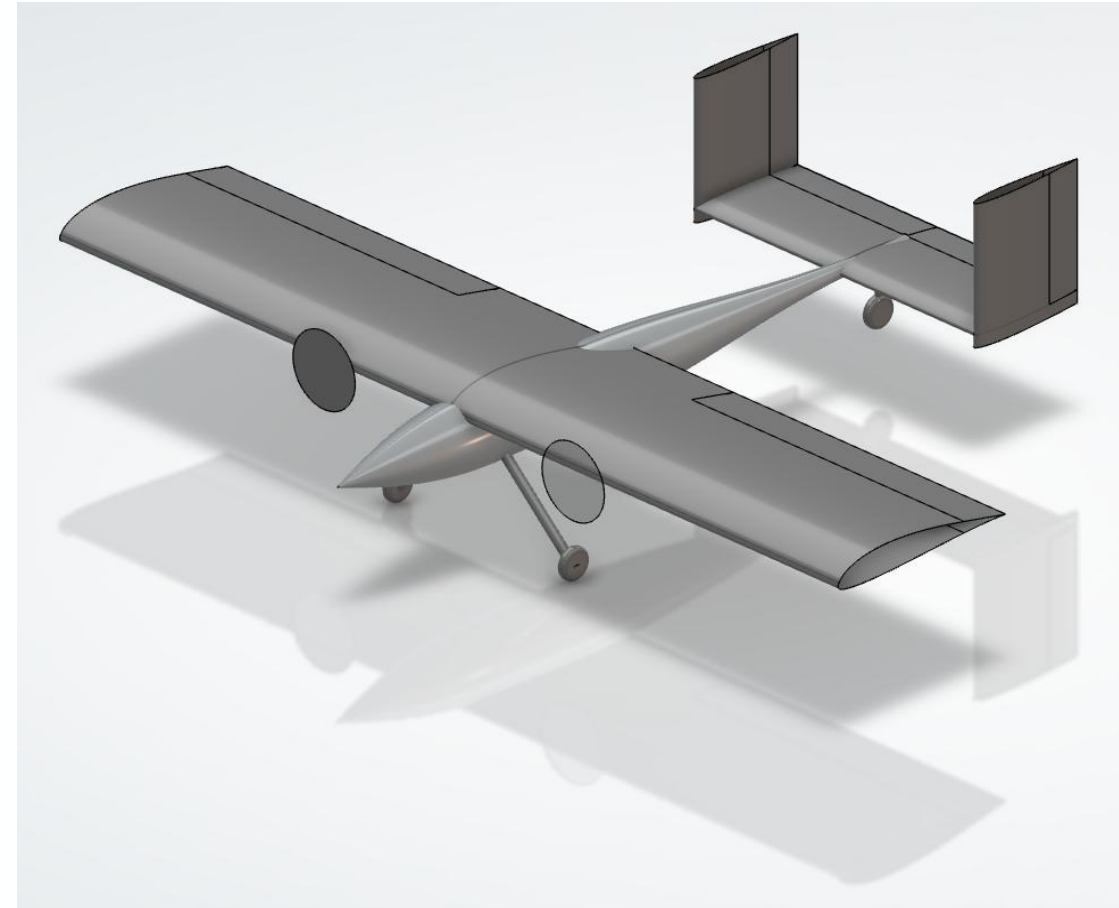
Wing Model

3DX Model – Updated with Wing Parameters

- This updated model in the team parameterized model is the **fifth row** in the configuration table.



Wing Model



Assembly Model